

PrefAgent – Preference Agent System
CAP4630 Project 3 – Spring 2026
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Introduction

This project implements a preference agent that models preference problems using binary attributes, hard constraints, and preference rules. The system supports both penalty logic and qualitative choice logic and reads input from files.

System Design

The program is divided into modules for parsing input, generating objects, evaluating constraints, and handling preference logic. Objects are represented as binary tuples corresponding to attribute values.

Implementation

All possible objects are generated and filtered based on hard constraints expressed in CNF. For penalty logic, penalties are assigned when rules are not satisfied and summed to determine optimal objects. For qualitative logic, objects are compared using dominance, and non-dominated objects are considered optimal.

Testing and Results

The system was tested using both the provided example and a custom test case with 8 attributes, 6 constraints, and multiple rules. This produced 256 total objects and 64 feasible ones. The program correctly performed encoding, feasibility checking, table generation, exemplification, and optimization for both logic types.

Conclusion

The system meets all project requirements and correctly implements preference reasoning using both penalty and qualitative approaches.

```
C:\Users\ethan\IdeaProjects\Project3\.venv\Scripts\python.exe C:\Users\ethan\IdeaProjects\Project3\main.py
Welcome to PrefAgent!
Enter Attributes File Name: attributes.txt
Enter Hard Constraints File Name: constraints.txt

Choose the preference logic to use:
1. Penalty Logic
2. Qualitative Choice Logic
3. Exit
Your Choice: 1
You picked Penalty Logic
Enter Preferences File Name: penaltylogic.txt

Choose the reasoning task to perform:
1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Your Choice: |
```

```
Choose the reasoning task to perform:
1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Your Choice: 1
o0 - ice-cream, beer, beef
o1 - ice-cream, beer, fish
o2 - ice-cream, wine, beef
o3 - ice-cream, wine, fish
o4 - cake, beer, beef
o5 - cake, beer, fish
o6 - cake, wine, beef
o7 - cake, wine, fish
```

```
Choose the reasoning task to perform:
1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Your Choice: 2
Yes, there are 6 feasible objects.
```

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Your Choice: 3

encoding	fish AND wine	wine OR cake	total penalty
o0	10	6	16
o1	10	6	16
o4	10	0	10
o5	10	0	10
o6	10	0	10
o7	0	0	0

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Your Choice: 4

Two randomly selected feasible objects are o4 and o1,
and o4 is strictly preferred over o1.

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Your Choice: 5

All optimal objects: o7

```

Choose the preference logic to use:
1. Penalty Logic
2. Qualitative Choice Logic
3. Exit
Your Choice: 2
You picked Qualitative Choice Logic
Enter Preferences File Name: qualitativechoicelogic.txt

```

```

Choose the reasoning task to perform:
1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Your Choice: 1
o0 - ice-cream, beer, beef
o1 - ice-cream, beer, fish
o2 - ice-cream, wine, beef
o3 - ice-cream, wine, fish
o4 - cake, beer, beef
o5 - cake, beer, fish
o6 - cake, wine, beef
o7 - cake, wine, fish

```

```

Choose the reasoning task to perform:
1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Your Choice: 2
Yes, there are 6 feasible objects.

```

```

Choose the reasoning task to perform:
1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu
Your Choice: 3

```

encoding	fish BT beef IF	ice-cream BT cake IF	wine BT beer IF	fish BT beer IF	wine BT beef IF
o0	2	1	-	1	
o1	1	1	2	-	
o4	2	2	-	1	
o5	1	2	2	-	
o6	2	2	-	2	
o7	1	2	1	-	

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Your Choice: 4

Two randomly selected feasible objects are o4 and o6,
and o4 is strictly preferred over o6.

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Your Choice: 5

All optimal objects: o0, o1, o7

Choose the reasoning task to perform:

1. Encoding
2. Feasibility Checking
3. Show the Table
4. Exemplification
5. Omni-optimization
6. Back to previous menu

Your Choice: 6

Choose the preference logic to use:

1. Penalty Logic
2. Qualitative Choice Logic
3. Exit

Your Choice: |